**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**
**MGU-UGP (HONOURS) REGULAR EXAMINATION ….**
**Fourth Semester**
MODEL QUESTION PAPER - THEORY
**Value Added Course - MG4VACSTA200–**
**ETHICAL DIMENSIONS IN STATISTICAL MACHINE LEARNING THROUGH R/PYTHON**
(2024 ADMISSION)

Duration: 1.5 Hours Maximum Marks: 50

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)**

*Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.*

[Learning Domain][CO No(s)]

**Part A**

Multiple Choice Questions

*Answer **All** Questions. Each question carries **1** mark*

1. Which primary data collection method is most suitable for illiterate respondents? [U][1]
   a. Questionnaire method b. Schedule method c. Online survey d. Email interview
2. When two variables move in the same direction, the correlation is: [U][1]
   a. Negative b. Zero c. Positive d. Perfect
3. Logistic regression predicts the probability of: [S][2]
   a. A continuous outcome b. A binary outcome c. Multiple outcomes simultaneously d. A time series
4. Which of the following is an example of a qualitative attribute? [U][1]
   a. Age of a person b. Gender of a person c. Income in dollars d. Height in cm
5. In Bayesian statistics, the prior probability represents: [S][2]
   a. Probability after seeing the data b. Probability based on past knowledge or belief
   c. Probability computed from the current sample d. Probability of the complement event
6. Which of the following is NOT a measure of variability? [U][1]
   a. Mean b. Standard deviation c. Variance d. Quartile deviation
7. Which of the following is considered a key principle of data ethics? [Ap][3]
   a. Transparency b. Privacy c. Accountability d. All of the above
8. Small sample tests are usually applied when the sample size (n) is: [U][1]
   a. $n > 30$ b. $n \leq 30$ c. $n > 50$ B. $n \geq 100$
9. In classification, the output variable is usually: [S][2]
   a. Continuous b. Categorical c. Random d. Unstructured
10. Which visualization is most suitable for comparing categories? [U][1]
    a. Line chart b. Bar chart c. Scatter plot d. Heat map

*(10x1=10 marks)*

**Part B**

Short Essay Questions

*Answer any **five** questions. Each question carries **6** marks*

11. Explain the ethical framework across the data analysis lifecycle. [Ap][3]
12. Demonstrate the difference between simple linear and logistic regression using examples. [S][2]
13. Under what conditions is a paired samples t-test used? In a study to test the impact of a new teaching method, paired t test was carried out and the output was obtained as t = 3.67, p=0.001. Illustrate the testing procedure and give the conclusion. [S][2]
14. Explain the ethical principles in data science. [Ap][3]
15. Distinguish between vectors, matrices, and data frames in R. Explain their structure and typical uses with suitable examples. [Ap][3]
16. Explain the concept of independence of attributes. [U][1]
17. Estimate mean and standard deviation of the following data

    5,10,20,15,9,6,12,8 [U][1]

*(5x6=30 marks)*

**Part C**

Essay Type Questions

*Answer any **one** question. Each question carries **10** marks*

18. Discuss the various methods of primary data collection. [U][1]
19. Explain the applications of classification models. Discuss the situations in which different classification algorithms such as KNN, Decision Tree, Random Forest, and SVM are most appropriate. [S][2]

*(1x10=10 marks)*